Influence of Cloudiness on the Performance of Cloud Computing

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Abstract: This paper is great. Just read it.

1 Introduction

Cloud computing is an important topic of our time. No one wants to run servers in the basement anymore, but everything should be scalable, cost-efficient and quickly calculated in the cloud. In the past the application itself was optimized to speed up calculation. This concerns topics such as horizontal scaling [4], utilization of data locality [5] and increased fault tolerance [6]. All of these things are important, but to the best of our knowledge, no one has tried to exploit fluctuations in the performance of the cloud itself. In this paper, we address the following hypothesis:

> "Cloud computing is more powerful when it is cloudy"

The hypothesis is intuitively easy to understand. If it is called cloud computing, then obviously clouds perform the calculations. If there are more clouds, the application has more computational power available and is therefore faster.

2 Related Work

This research is completely novel and there is no related work. We boldly go where no man has gone before.

3 Approach

To prove our hypothesis, we ran multiple CPU benchmarks under different weather conditions. Unfortunately, public clouds were not suitable for this experiment because their exact location is often unknown. This prevented us from monitoring the actual weather at the time of the benchmarks.

To solve the problem, we simulated our own cloud using a Raspberry Pi 5 (4GB) running Ubuntu 24.04. We placed the device in our garden with a direct line of sight to the sky and equipped it with rain protection (see Fig. 1).

To measure the performance of our cloud, we ran SysBench [1] (version 1.0.20) every 15 minutes. SysBench counted how often it could calculate the prime numbers up to 10,000 on 2 threads. We ran the test for 10 minutes and recorded the events per second.

```
sysbench --threads=2 --time=600 cpu run
```

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(a) Test Setup (b) Rain Protection

Figure 1: Our test setup consists of a Raspberry Pi to simulate a cloud environment and a webcam to document cloud coverage in the sky. The ESP32 and the Brightness Sensor were installed but not used.

To prove our hypothesis, we also needed measurements of cloud coverage. For this, we installed an ESP32 with a brightness sensor next to the Raspberry Pi. The cloudier it is, the darker it gets. Together with the expected brightness based on the position of the sun, we wanted to estimate the cloud coverage. However, the used brightness sensor was too inaccurate. Our measurements were almost binary: It was dark during night and bright during day. This made it impossible to derive a continuous spectrum of cloud density.

Therefore, we added a webcam to the Raspberry Pi to take pictures of the sky. The major drawback of this solution is that we have to classify the images manually. (I have a bad feeling about this. But it would be a great task for some student assistants[1].) To minimize classification work, we reduced the image capture frequency to one picture every 20 minutes during the working time between 9:00 AM and 5:00 PM. There are some solutions for automated cloud detection using all-sky imagers [3, 7]. However, they use fisheye lenses and we don't like fish. Other solutions, such as NubiScope [2], are too complex, even for such an important paper.

Our final architecture is shown in Fig. 2. It consists of the Raspberry Pi, which runs the benchmark every 15 minutes, and the webcam, which takes a sky photo every 20 minutes. The results are stored in an Influx database on a remote server and in Amazon S3. (This is nice because S3 is a cloud environment that is used to store the cloud environment - we love recursion :exploding-head-emoji)

[1] No students were harmed in the making of this paper.

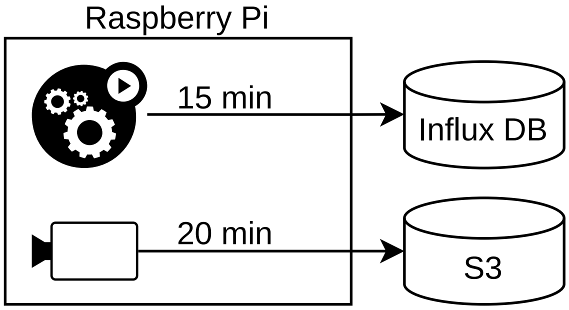


Figure 2: Our final architecture. We run a CPU benchmark every 15 minutes to measure cloud computing performance. Every 20 minutes, a webcam takes photos of the sky to document cloudiness.

4 Results

We performed our measurements between the 17th of June 2025 and the 2nd of November 2025 in Hesse, Germany [8]. In total, we ran 4,015 benchmarks during working hours (9 a.m. to 5 p.m.) and combined them with 2,977 photos of the sky. On some days, we had network problems or system crashes, resulting in gaps in the time series – who cares...

We manually classified all images of the sky into five groups depending on their level of cloud coverage: 100%, 75%, 50%, 25% and no clouds in the sky. Examples are shown in Fig. 3. In total we had 1,079 photos with 100% clouds, 476 with 75%, 457 with 50%, 393 with 25% and 572 with no clouds at all.

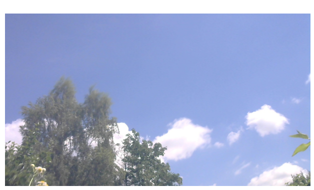
We combined each benchmark result with the closest photo and grouped the pairs by cloud coverage level. On the resulting sets we were able to determine the performance of our cloud as visualized in Fig. 4.

When there is no cloud coverage, there is a wide range of benchmark results, with the median at 4,465 points. 4,465 is a large number, so cloud computing itself is very good.

(a) 100% Clouds

(b) 75% Clouds

(c) 50% Clouds



(d) 25% Clouds

(e) No Clouds

Figure 3: Example for different cloud coverage levels captured by the webcam. Classification was done manually.

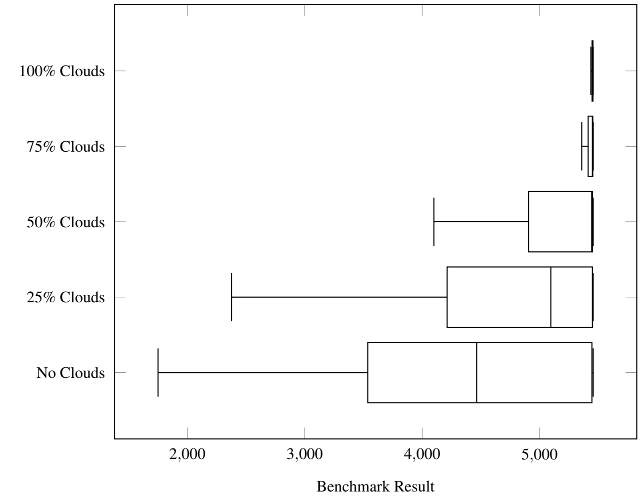


Figure 4: Benchmark results for different cloud coverage levels. Higher is better.

However, as cloud coverage increases, it gets even better. Already 25% cloud coverage leads to a median of 5,096 points, and from 50% onward, the maximum value of approximately 5,444 points is reached. This is an increase of about 22% compared to the median with no clouds. Further increases in cloud coverage only reduce the variance in benchmark results. When the sky is completely covered with clouds, you can almost be sure to realize the maximum possible performance. This is great!

5 Conclusion & Future Work

In this paper, we showed that cloud coverage in the sky has a direct impact on the performance of cloud computing. On cloudy days, performance can be increased by approximately 22%. We are still unsure how our cloud on the ground communicates with the clouds in the sky. It probably has something to do with quantum physics, AI and blockchain.

We also want to investigate whether the shape of clouds has an influence on performance. Presumably, turtle clouds are significantly slower than leopard clouds. A future study should take a closer look at this.

The same applies to the use of rainwater. We know that clouds consist of water vapor. After condensation the rain probably contains all stored user data. So be careful: If you don't encrypt your data, anyone with a rain barrel can perform data mining on them. But you can also exploit this for cloud migrations. Don't pay for outgoing traffic, just wait for rain.

Finally, we want to emphasize that our measured performance increase was not caused by other effects than the presence of real clouds. An unlikely idea would be that the Raspberry Pi heats up to over 80°C in direct sunlight and therefore throttles down. If the duration of the benchmarks, the number of threads and the focus on the time between 9:00 AM and 5:00 PM had been deliberately chosen to cause this effect, it would be an inappropriate manipulation of the study. We therefore distance ourselves from any accusations in this regard. Live long and prosper!

REFERENCES

[1] akopytov/sysbench: Scriptable database and system performance benchmark. https://github.com/akopytov/sysbench. Accessed: 2025-12-27.

[2] Nubiscope - overview. http://www.nubiscope.eu/index.html. Accessed: 2025-12-24.

[3] J. Esteves, Y. Cao, N. P. da Silva, R. Pestana, and Z. Wang. Identification of clouds using an all-sky imager. In *2021 IEEE Madrid PowerTech*, pages 1–5. IEEE, 2021.

[4] Q. Jiang, Y. C. Lee, and A. Y. Zomaya. The limit of horizontal scaling in public clouds. *ACM Transactions on Modeling and Performance Evaluation of Computing Systems (TOMPECS)*, 5(1):1–22, 2020.

[5] J. Jin, J. Luo, A. Song, F. Dong, and R. Xiong. Bar: An efficient data locality driven task scheduling algorithm for cloud computing. In *2011 11th IEEE/ACM International Symposium on Cluster, Cloud and Grid Computing*, pages 295–304. IEEE, 2011.

[6] P. Kumari and P. Kaur. A survey of fault tolerance in cloud computing. *Journal of King Saud University-Computer and Information Sciences*, 33(10):1159–1176, 2021.

[7] B. Nouri, P. Kuhn, S. Wilbert, N. Hanrieder, C. Prahl, L. Zarzalejo, A. Kazantzidis, P. Blanc, and R. Pitz-Paal. Cloud height and tracking accuracy of three all sky imager systems for individual clouds. *Solar Energy*, 177:213–228, 2019.

[8] H. M. Würz. Photos of the sky in 2025 combined with cpu benchmarks, 2026. doi:10.5281/zenodo.18722735.